Software Test Plan (STP) - Blood Bank Management System (BBMS)

Project: Blood Bank Management System (BBMS)
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1. Introduction

Purpose: This document defines the test plan for the Blood Bank Management System (BBMS) v1.0. It outlines the test objectives, scope, strategy, security validation, resources, schedule, test cases and responsibilities needed to verify the system against the SRS.

Scope: Testing covers all functional and non-functional requirements in SRS v1.0: user registration and login, admin management, donor management and eligibility, search of blood stock and donors, blood donation recording, blood stock management, blood request handling and reports, together with the performance, safety, security and quality requirements. Third-party infrastructure internals are excluded.

References: BBMS SRS v1.0 (02-10-2026); BBMS Software Architecture & Design Specification (SAD) v1.0 (in preparation); IEEE Std 829-1998, Standard for Software Test Documentation; Digital Personal Data Protection Act, 2023 (India).

Definitions: BBMS (Blood Bank Management System), STP (Software Test Plan), SRS (Software Requirements Specification), SAD (Software Architecture & Design Specification), RTM (Requirements Traceability Matrix), FEFO (First Expiry, First Out), RBAC (Role-Based Access Control), TC (Test Case), FR / NFR (Functional / Non-Functional Requirement), SO (Security Objective), BR (Business Rule).

2. Test Items

- User registration, login and profile module (SRS 5.1)
- Admin management and audit log module (SRS 5.2)
- Donor management and eligibility engine (SRS 5.3)
- Stock and donor search module (SRS 5.4)
- Blood donation recording and health screening module (SRS 5.5)
- Blood stock management and automated expiry job (SRS 5.6)
- Blood request handling and FEFO reservation module (SRS 5.7)
- Reports and PDF/CSV export module (SRS 5.8)
- REST API layer, role-based access control and database

3. Features to be Tested

Features mapped to SRS requirement IDs:

Feature area	SRS requirement IDs	What is verified
User registration and login (5.1)	FR-AUTH-01 to FR-AUTH-07	Registration with validation, duplicate rejection, login by role, 15-minute lockout, password reset, profile update, logout and session end
Admin management (5.2)	FR-ADM-01 to FR-ADM-06	Staff account creation, user list and filters, activate/deactivate, role assignment, audit log, last-admin protection
Donor management (5.3)	FR-DON-01 to FR-DON-08	Add/update/deactivate donors, data validation, eligibility rules and reasons, donor history, account linking
Search blood and donors (5.4)	FR-SRCH-01 to FR-SRCH-06	Stock search, result details, donor search, access denied for other roles, no-result message, pagination
Blood donation recording (5.5)	FR-DNR-01 to FR-DNR-06	Donation entry, eligibility check, blood unit creation and expiry, failed screening, last-donation update, 24-hour correction
Blood stock management (5.6)	FR-STK-01 to FR-STK-07	Stock by group and status, shelf life, midnight expiry, low-stock and expiring-soon alerts, discard, non-negative stock
Blood request handling (5.7)	FR-REQ-01 to FR-REQ-08	Submit, validate, Pending queue order, approve with FEFO, reject, issue, track and cancel
Reports (5.8)	FR-RPT-01 to FR-RPT-06	Stock, Donation and Request reports, filters, CSV/PDF export, access control, empty result
Performance	NFR-PERF-01 to NFR-PERF-04	Search and page load 3 s (95%), 50 concurrent users, report within 10 s
Safety	NFR-SAF-01 to NFR-SAF-03	No unsafe issue, blood group confirmation, daily backup with 30-day retention
Security	NFR-SEC-01 to NFR-SEC-07 (SO-1 to SO-4)	Password policy and hashing, RBAC, HTTPS, session security, input validation, audit trail, no personal data in logs
Software quality	NFR-QUAL-01 to NFR-QUAL-05	Usability, 99% availability, atomic transactions, modularity and 70% unit-test coverage, browser compatibility
Business rules	BR-01 to BR-06	Role limits, own-data access, eligibility, FEFO, urgent-first, staff-only approval

4. Features Not to be Tested

- SMS or push notifications and email/SMS reminders to donors and requesters (out of scope for v1.0)
- Online payments and live location tracking (out of scope for v1.0)
- Automatic blood group compatibility substitution and blood component separation (out of scope for v1.0)
- Mobile application (future enhancement)
- Internals of the SMTP email service, operating system and database engine (only the application integration is tested)
- Physical blood handling, laboratory screening procedures and identity checks done outside the system

5. Test Approach / Strategy

Levels:

- Unit tests: business logic such as eligibility calculation, expiry date calculation and stock decrement (Jest)
- Integration tests: REST API endpoints, HTTP status codes, JSON schemas, and the application with the database (Supertest, Postman)
- System tests: end-to-end workflows through the browser, across Chrome, Firefox and Edge
- Acceptance tests (UAT): Product Owner verifies each feature against its SRS requirement IDs

Types:

- Functional testing (all FR and business rules)
- Regression testing after each sprint and after defect fixes
- Performance testing (response time, 50 concurrent users, 10,000 donors and 50,000 units) with JMeter
- Usability testing (3 first-time staff users, alerts that do not rely on colour alone)
- Security testing (see Section 5.1)

Entry Criteria: Stable build delivered for the sprint; SRS v1.0 and SAD baselined; test environment ready with seed test data.

Exit Criteria: 100% of planned test cases (TC-01 to TC-37) executed; 0 critical or high defects open; all acceptance criteria satisfied; RTM complete.

5.1 Security Validation

Validation procedures for Security Objectives SO-1 to SO-4 and requirements NFR-SEC-01 to NFR-SEC-07:

Security area	SRS ref	Validation procedure	Test case
Password policy and hashing	NFR-SEC-01 (SO-1)	Verify the policy (8+ characters, upper, lower, digit); confirm passwords are stored only as salted bcrypt hashes and never logged.	TC-01
Role-based access control	NFR-SEC-02 (SO-1, SO-2)	Access admin, donor search and report endpoints as Donor and Hospital/Patient; verify HTTP 403 and that the attempt is logged.	TC-09, TC-24
Transport encryption	NFR-SEC-03 (SO-1)	Verify HTTP redirects to HTTPS and only TLS 1.2 or higher is accepted.	TC-35
Session security	NFR-SEC-04 (SO-1, SO-3)	Verify 15-minute inactivity timeout; check HttpOnly and Secure flags on session cookies.	TC-18
Input sanitisation and injection	NFR-SEC-05 (SO-2)	Submit SQL payloads (' OR 1=1 --) and script tags in all text inputs; verify parameterised queries and rejection.	TC-16
Audit trail	NFR-SEC-06 (SO-4)	Perform logins, role changes, donations and request decisions; verify user ID, timestamp and action are recorded and kept 1 year.	TC-21
No personal data in logs	NFR-SEC-07 (SO-1)	Inspect application and error logs and error messages for passwords, phone numbers, emails and addresses.	TC-35
Availability and lockout abuse	SO-3, FR-AUTH-04	Repeated failed logins lock only the targeted account for 15 minutes; other users are unaffected.	TC-02
Fuzzing of input fields	NFR-SEC-05	Send malformed, oversized and special-character values to registration, donor, request and search forms and APIs; verify safe validation errors.	TC-13, TC-16, TC-22
Penetration test of authentication flows	NFR-SEC-02, NFR-SEC-04	Attempt session reuse after logout, URL tampering to read another user's data, and direct API calls without a token.	TC-09, TC-18, TC-24

6. Test Environment

Hardware: Developer laptops or a test server (4 GB RAM or more); client machines on Windows, Linux or macOS, and an Android phone for mobile-width checks.

Software: BBMS web application v1.0 (Node.js with REST API and JSON, as decided for the SAD); relational database (PostgreSQL or MySQL as finalised in the SAD); browsers Chrome, Firefox and Edge (latest two versions); SMTP test mailbox for password reset links.

Tools: Jest and Supertest (unit and API tests), Postman (API), JMeter (performance), Git/GitHub (version control), GitHub Issues or Jira (defect tracking), browser developer tools and an HTTPS scanner (security checks).

Test Data: Seed accounts for each role (Donor, Hospital/Patient, Staff, Admin); sample donors covering eligible and ineligible cases (age, weight, gender, gap); blood units with different expiry dates for all 8 blood groups; a bulk dataset of 10,000 donors and 50,000 units for performance tests.

7. Test Schedule

Milestones (dates to be aligned with the sprint plan):

- Test case design: 05-10-2026
- Test Plan review and sign-off: 06-10-2026; upload to repository: 07-10-2026
- Environment and seed data setup: 14-10-2026
- Test execution cycle 1 (authentication, admin, donor): 19-10-2026 to 23-10-2026
- Test execution cycle 2 (donation, stock, requests, reports): 02-11-2026 to 06-11-2026
- Regression, security and performance testing: 09-11-2026 to 13-11-2026
- UAT and Test Summary Report: 16-11-2026 to 18-11-2026

8. Test Deliverables

- Test Plan (this document)
- Test Cases (manual and automated), Section 13
- Test Scripts (Jest, Supertest, Postman collections, JMeter plans)
- Test Data (seed and bulk datasets)
- Test Execution Logs
- Defect Reports
- Requirements Traceability Matrix (Section 13.2)
- Test Summary Report

9. Roles and Responsibilities

Role	Name	Responsibility
QA Lead	Dheepak Seetharam	Prepare plan, coordinate execution
Test Engineer	Chukkapalli Rohan	Design & execute test cases, log defects
Developer	Gautam Prasanth Nair	Support defect fixes and triage
Product Owner	Harsimran Kaur	Approve test results, sign-off readiness

10. Risks and Mitigation

Risk	Mitigation
Delay in stable build delivery	Request early smoke builds each sprint; test the API contracts first

Risk	Mitigation
Technology stack or database choice still changing	Keep test cases stack-neutral and driven by SRS IDs; update the tool section once the SAD is final
Dependency on the SMTP/email service for password reset	Use a test mailbox and a mock SMTP stub so reset tests do not depend on the live service
Database schema changes	Keep version-controlled seed and migration scripts so test data can be rebuilt quickly
Tight project timeline with a team of four	Run high-priority security and functional test cases first; automate regression tests
Test environment downtime	Keep a local Docker or backup environment and reset scripts

11. Assumptions & Dependencies

- The approved SRS v1.0 is the source of truth for expected results
- Test data can be reset between test cycles
- An email service (or test mailbox) is available for password reset links
- Eligibility limits, shelf life and low-stock threshold use the SRS default values during testing
- Features are delivered per the sprint plan so each test cycle has a stable build

12. Suspension & Resumption Criteria

Suspend testing if:

- The test environment is unavailable for more than 4 hours
- The build is too unstable and blocks more than 30% of test cases, or the application cannot start

Resume testing if:

- Blocking defects are resolved and a corrected build passes smoke testing
- The environment and database connectivity are stable again

13. Test Case Management & Traceability

The RTM (13.2) maps every SRS requirement (all FR, NFR and business rules) to one or more test cases. The 37 test cases below cover both functional and non-functional requirements.

13.1 Test Cases

TC ID	Test case description	SRS reference	Test steps / inputs	Expected result	Type / priority
TC-01	Valid User Registration & Password Hashing	FR-AUTH-01 NFR-SEC-01	1. Submit registration with valid name, email, phone and a strong password. 2. Inspect the User table record. 3. Try password "abc123".	Account created; password stored only as salted bcrypt hash; weak password rejected with a message.	Functional / High
TC-02	Account Lockout on Repeated Failed Logins	FR-AUTH-04	1. Enter valid email with wrong password 5 consecutive times. 2. Then enter the correct password.	Account locked for 15 minutes; remaining lock time shown; correct password also refused during lock.	Security / High
TC-03	Donor Eligibility Gender Gap Rule	FR-DON-04 BR-03	1. Check a Male donor whose last donation was 100 days ago. 2. Check a Female donor whose last donation was 100 days ago.	Male shown "Eligible" (100 d >= 90 d); Female shown "Not eligible" (100 d < 120 d) with next eligible date.	Functional / High
TC-04	Blood Donation Entry & Automatic 35-Day Expiry	FR-DNR-01 FR-DNR-03	1. Staff select an eligible B+ donor (blood group auto-filled). 2. Enter collection date 04-10-2026, volume 350 ml, screening Passed. 3. Submit.	Donation recorded; unit UNT-XXXXX created with status Available and expiry date 08-11-2026 (04-10-2026 + 35 days).	Functional / High
TC-05	Failed Health Screening Handling	FR-DNR-04	1. Staff record a donation with Screening Status = Failed.	Donation saved as Failed Screening; NO blood unit created; reason logged in donor medical notes.	Functional / Medium
TC-06	Midnight Automated Stock Expiry	FR-STK-03 NFR-SAF-01	1. Set system clock to 00:00 on the expiry date of an Available unit. 2. Let the daily job run. 3. Try to issue that unit.	Unit status changes from Available to Expired; unit is excluded from stock search and blocked from issue.	Functional / High
TC-07	Low-Stock Alert Badge Triggering	FR-STK-04	1. Reduce available O- units to 4 (below threshold of 5). 2. Open the staff dashboard.	Dashboard shows a warning with text "Low Stock Alert: O-" (colour is not the only signal).	Functional / High
TC-08	Blood Request Approval & FEFO Reservation	FR-REQ-05 BR-04	1. Staff approve a request for 2 units of A+ when 4 A+ units with different expiry dates exist.	The 2 units with the earliest expiry dates are set to Reserved; request status becomes Approved.	Functional / High
TC-09	Unauthorized Access Block (RBAC Enforcement)	FR-SRCH-04 FR-RPT-06 NFR-SEC-02	1. Log in as Hospital/Patient. 2. Call the donor search and report endpoints (UI and API).	Access refused with HTTP 403 "access denied"; attempt logged.	Security / High
TC-10	Report Generation & PDF/CSV Export	FR-RPT-01 FR-RPT-05	1. Staff open Stock Report for a date range. 2. Click Export PDF. 3. Click Export CSV.	Report shows available, reserved, issued, expired, discarded units per blood group and earliest expiry; PDF and CSV files download with the same figures.	Functional / Medium
TC-11	Admin User Deactivation & Last Admin Protection	FR-ADM-03 FR-ADM-06	1. Admin deactivates a staff account; that user tries to log in. 2. Admin tries to deactivate the last active admin.	Deactivated user cannot log in; deactivating the last active admin is blocked with an error message.	Functional / High

TC ID	Test case description	SRS reference	Test steps / inputs	Expected result	Type / priority
TC-12	Stock Search Filtering & No-Result Handling	FR-SRCH-01 FR-SRCH-02 FR-SRCH-05	1. Hospital user searches stock for a blood group with 0 units. 2. Search a group that has units (as requester, then as staff).	Zero results show "No matching results found" with the criteria and no error; results list only Available, unexpired units; requester sees count and earliest expiry, staff also see Unit IDs; no donor identity shown.	Functional / High
TC-13	Request Parameter Validation Rejection	FR-REQ-02	1. Submit a request with units = 11. 2. Submit with a past required-by date. 3. Submit with a 9-digit contact number.	Each request is rejected with a specific error message next to the invalid field; nothing is saved.	Functional / Medium
TC-14	Priority Request Queue Ordering	FR-REQ-04 BR-05	1. Submit a Normal request. 2. Then submit an Urgent request. 3. Open the staff pending queue.	Urgent request is listed before the Normal request; same priority sorted by oldest first.	Functional / High
TC-15	Block Unsafe Unit Issuance & Group Mismatch	NFR-SAF-01 NFR-SAF-02 FR-REQ-07	1. Try to issue an Expired unit, a Discarded unit and a unit reserved for another request. 2. Try to issue an O+ unit for an A+ request. 3. Issue a correct unit and confirm the group match.	Unsafe issues blocked with an error message; staff must confirm blood group match before the request can be marked Issued; unit statuses become Issued.	Safety / High
TC-16	SQL Injection & Script Payload Rejection	NFR-SEC-05	1. Enter ' OR 1=1 -- in login and search fields. 2. Enter <script>alert(1)</script> in text inputs.	Inputs are sanitised and handled with parameterised queries; attempts rejected without data exposure or stack trace.	Security / High
TC-17	Search Performance & Concurrency	NFR-PERF-01 NFR-PERF-03	1. Load 10,000 donors and 50,000 units. 2. Run search queries. 3. Simulate 50 concurrent logged-in users (JMeter).	95% of searches return within 3 seconds; average response time with 50 users is no more than 3 seconds.	Performance / Medium
TC-18	Session Expiry & Cookie Security Flags	FR-AUTH-07 NFR-SEC-04	1. Log in and stay idle for 15 minutes, then act. 2. Log out and press Back. 3. Inspect the session cookie.	Session ends after 15 minutes of inactivity and after logout; login required again; cookie marked HttpOnly and Secure.	Security / High
TC-19	Duplicate Account Rejection & Role-Based Login	FR-AUTH-02 FR-AUTH-03 NFR-PERF-02	1. Register with an email already in use; repeat with an existing phone number. 2. Log in separately as Donor, Hospital/Patient, Staff and Admin and measure page load.	"Account already exists" shown for both; each role lands on its own dashboard within 3 seconds (95% of loads) showing only its permitted menu items.	Functional / High
TC-20	Password Reset & Profile Update	FR-AUTH-05 FR-AUTH-06	1. Request a reset and use the one-time link within 15 minutes. 2. Reuse the same link. 3. Use a link older than 15 minutes. 4. As a donor, update name, phone and address.	First use succeeds; reused and expired links are rejected; profile changes are saved and displayed.	Functional / Medium
TC-21	Admin Account Management & Audit Log	FR-ADM-01 FR-ADM-02 FR-ADM-04 FR-ADM-05 NFR-SEC-06 BR-01	1. Admin creates a staff account (name, email, phone, temporary password). 2. Filter the user list by role and by status. 3. Change a user's role. 4. Open the audit log as Admin, then as Staff.	Staff account created; filters return matching users; role change applies immediately and is logged with user, action and timestamp; Staff cannot open the audit log (403).	Functional / High

TC ID	Test case description	SRS reference	Test steps / inputs	Expected result	Type / priority
TC-22	Add Donor & Field Validation	FR-DON-01 FR-DON-02 BR-01	1. Staff add a donor with all valid mandatory fields. 2. Retry with weight 20 kg, 9-digit phone, 5-digit PIN, future date of birth. 3. Add a donor with an existing phone number.	Valid donor saved with unique Donor ID and status Active; each invalid field shows an error beside it; duplicate rejected with "Donor already exists".	Functional / High
TC-23	Update Donor & Ineligibility Reasons	FR-DON-03 FR-DON-05	1. Staff edit a donor's city and weight (try editing Donor ID). 2. Open the record of a donor weighing 44 kg whose last donation was 40 days ago.	Update saved with user and timestamp, donation history unchanged, Donor ID not editable; record lists every failed criterion and the next eligible date.	Functional / Medium
TC-24	Donor History & Account Linking	FR-DON-06 FR-DON-08 BR-02	1. Log in as a donor whose phone matches a donor record with 3 donations. 2. Log in as a donor with no matching record. 3. Change the donor ID in the URL to another donor's.	Only own donations shown, newest first, with Donation ID, date, blood group and volume; unlinked donor sees "No donations recorded yet"; other donor's data refused (403).	Functional / High
TC-25	Deactivate Donor, No Deletion	FR-DON-07	1. Staff deactivate a donor. 2. Search donors by that blood group. 3. Try recording a donation for them. 4. Look for a delete option or call a delete API. 5. Reactivate the donor.	Donor is Inactive, absent from search results, cannot be recorded for donation; history retained; no deletion possible; reactivation restores normal behaviour.	Functional / Medium
TC-26	Donor Search & Pagination	FR-SRCH-03 FR-SRCH-06	1. Staff search donors by blood group O+ with 45 matching eligible donors; add city or PIN. 2. Move to next and previous page.	Only Active, Eligible donors listed with Donor ID, name, blood group, city, phone and last donation date; 20 per page; total (45) and next/previous controls shown.	Functional / Medium
TC-27	Donation Eligibility Block, Last-Donation Update & 24-Hour Correction	FR-DNR-02 FR-DNR-05 FR-DNR-06 BR-03	1. Try recording a donation for a Not Eligible donor. 2. Record a donation for an eligible donor and check Last Donation Date and next eligible date. 3. Edit that donation's volume within 24 hours. 4. Attempt the same edit after 24 hours.	Ineligible submission blocked with the exact reason; last donation and next eligible dates updated; edit within 24 h logged with staff ID, timestamp, original and new value; edit after 24 h refused.	Functional / High
TC-28	Stock Dashboard, Shelf Life & Expiring Soon	FR-STK-01 FR-STK-02 FR-STK-05	1. Open the stock dashboard. 2. Add units expiring in 6, 7 and 8 days. 3. Open the unit list for one blood group.	Counts for Available, Reserved, Issued, Expired and Discarded shown for all 8 blood groups; default expiry = collection date + 35 days; units within 7 days tagged "Expiring Soon" with text.	Functional / Medium
TC-29	Discard Unit & Negative-Stock Guard	FR-STK-06 FR-STK-07	1. Discard a unit without selecting a reason. 2. Discard a unit with reason "Hemolyzed". 3. Try approving a request for more units than are Available.	Discard without reason blocked; valid discard sets status Discarded and logs reason and staff ID; allocation beyond stock refused and counts never go negative.	Functional / High
TC-30	Submit, Track & Cancel Blood Request	FR-REQ-01 FR-REQ-03 FR-REQ-08 BR-02	1. Hospital user submits a valid request (patient, group, 2 units, date, hospital, contact, priority). 2. Open "My requests". 3. Cancel while Pending. 4. Try to cancel an Approved request. 5. Log in as another requester.	Unique Request ID created with status Pending and shown in staff queue; user sees only own requests; cancel works only while Pending; other users' requests are not visible.	Functional / High
TC-31	Reject Request & Approval Authority	FR-REQ-06 BR-06	1. Staff reject a Pending request with a blank reason, then with a reason. 2. Hospital user tries to approve or reject (UI and API).	Blank reason refused; rejection saves status Rejected, reason, staff ID and timestamp, and the reason is shown to the requester; requester attempt returns 403.	Functional / High

TC ID	Test case description	SRS reference	Test steps / inputs	Expected result	Type / priority
TC-32	Donation & Request Reports with Filters	FR-RPT-02 FR-RPT-03 FR-RPT-04 NFR-PERF-04	1. Generate Donation and Request reports for a date range filtered by blood group and status. 2. Generate a report covering 12 months of data. 3. Apply filters that match no data.	Reports show all required fields and totals; 12-month report appears within 10 seconds; empty result shows "No records found".	Functional / Medium
TC-33	Atomic Transactions	NFR-QUAL-03	1. Inject a failure (e.g., stop the database connection) midway through recording a donation, a discard and a request approval. 2. Check donation, blood unit and stock data.	Each operation is either fully saved or fully rolled back; stock and records never go out of step.	Reliability / High
TC-34	Backup & Restore	NFR-SAF-03	1. Confirm an automatic backup runs every 24 hours and backups are kept 30 days. 2. Restore the latest backup into a clean database.	Daily backup exists with 30-day retention; restore recovers the data with at most one day lost.	Safety / Medium
TC-35	TLS Enforcement & No Personal Data in Logs	NFR-SEC-03 NFR-SEC-07	1. Open the site over HTTP and scan the supported TLS versions. 2. Trigger errors (bad login, invalid donor) and inspect application logs, error logs and on-screen messages.	HTTP redirects to HTTPS and only TLS 1.2 or higher is accepted; no passwords, phone numbers, emails or addresses appear in logs or error messages.	Security / High
TC-36	Usability & Browser Compatibility	NFR-QUAL-01 NFR-QUAL-05	1. Three first-time staff users register a donor and record a donation using only on-screen guidance. 2. Open the main screens in the latest two versions of Chrome, Firefox and Edge at 360 px width.	Each user completes the task within 5 minutes; the UI works and is usable at 360 px width in all three browsers.	Usability / Medium
TC-37	Availability & Unit-Test Coverage	NFR-QUAL-02 NFR-QUAL-04	1. Monitor uptime over a calendar month (excluding announced maintenance). 2. Run the unit test suite with a coverage report.	Availability is at least 99%; code is organised in per-feature modules and unit tests cover at least 70% of business logic.	Quality / Medium

13.2 Requirements Traceability Matrix (RTM)

Mapping of SRS v1.0 requirements to test cases. The Architecture and Design reference columns are completed from the SAD once it is finalised.

Sl.	Requirement ID	Description	Test case ID	Test status	Architecture ref	Design ref
1	FR-AUTH-01	Register account	TC-01	Planned	See SAD	See SAD
2	FR-AUTH-02	Reject duplicate account	TC-19	Planned	See SAD	See SAD
3	FR-AUTH-03	Login by role	TC-19	Planned	See SAD	See SAD
4	FR-AUTH-04	Account lockout	TC-02	Planned	See SAD	See SAD
5	FR-AUTH-05	Password reset	TC-20	Planned	See SAD	See SAD
6	FR-AUTH-06	Manage own profile	TC-20	Planned	See SAD	See SAD
7	FR-AUTH-07	Logout and session end	TC-18	Planned	See SAD	See SAD
8	FR-ADM-01	Create staff account	TC-21	Planned	See SAD	See SAD
9	FR-ADM-02	View and filter users	TC-21	Planned	See SAD	See SAD
10	FR-ADM-03	Activate / deactivate user	TC-11	Planned	See SAD	See SAD

Sl.	Requirement ID	Description	Test case ID	Test status	Architecture ref	Design ref
11	FR-ADM-04	Assign role	TC-21	Planned	See SAD	See SAD
12	FR-ADM-05	Audit log	TC-21	Planned	See SAD	See SAD
13	FR-ADM-06	Protect last admin	TC-11	Planned	See SAD	See SAD
14	FR-DON-01	Add donor record	TC-22	Planned	See SAD	See SAD
15	FR-DON-02	Validate donor data	TC-22	Planned	See SAD	See SAD
16	FR-DON-03	Update donor and history	TC-23	Planned	See SAD	See SAD
17	FR-DON-04	Eligibility check	TC-03	Planned	See SAD	See SAD
18	FR-DON-05	Show ineligibility reason	TC-23	Planned	See SAD	See SAD
19	FR-DON-06	Donor views own history	TC-24	Planned	See SAD	See SAD
20	FR-DON-07	Deactivate, not delete	TC-25	Planned	See SAD	See SAD
21	FR-DON-08	Link donor to account	TC-24	Planned	See SAD	See SAD
22	FR-SRCH-01	Search stock by group	TC-12	Planned	See SAD	See SAD
23	FR-SRCH-02	Stock result details	TC-12	Planned	See SAD	See SAD
24	FR-SRCH-03	Search donors	TC-26	Planned	See SAD	See SAD
25	FR-SRCH-04	Hide donor details	TC-09	Planned	See SAD	See SAD
26	FR-SRCH-05	No result message	TC-12	Planned	See SAD	See SAD
27	FR-SRCH-06	Paginate results	TC-26	Planned	See SAD	See SAD
28	FR-DNR-01	Record donation	TC-04	Planned	See SAD	See SAD
29	FR-DNR-02	Block ineligible donor	TC-27	Planned	See SAD	See SAD
30	FR-DNR-03	Create blood unit	TC-04	Planned	See SAD	See SAD
31	FR-DNR-04	Failed screening	TC-05	Planned	See SAD	See SAD
32	FR-DNR-05	Update last donation	TC-27	Planned	See SAD	See SAD
33	FR-DNR-06	Correct entry	TC-27	Planned	See SAD	See SAD
34	FR-STK-01	Stock by group and status	TC-28	Planned	See SAD	See SAD
35	FR-STK-02	Expiry date	TC-28	Planned	See SAD	See SAD
36	FR-STK-03	Auto-expire units	TC-06	Planned	See SAD	See SAD
37	FR-STK-04	Low-stock alert	TC-07	Planned	See SAD	See SAD
38	FR-STK-05	Expiry alert	TC-28	Planned	See SAD	See SAD
39	FR-STK-06	Discard unit	TC-29	Planned	See SAD	See SAD
40	FR-STK-07	No negative stock	TC-29	Planned	See SAD	See SAD

Sl.	Requirement ID	Description	Test case ID	Test status	Architecture ref	Design ref
41	FR-REQ-01	Submit request	TC-30	Planned	See SAD	See SAD
42	FR-REQ-02	Validate request	TC-13	Planned	See SAD	See SAD
43	FR-REQ-03	Assign ID, set Pending	TC-30	Planned	See SAD	See SAD
44	FR-REQ-04	Pending queue order	TC-14	Planned	See SAD	See SAD
45	FR-REQ-05	Approve and reserve	TC-08	Planned	See SAD	See SAD
46	FR-REQ-06	Reject with reason	TC-31	Planned	See SAD	See SAD
47	FR-REQ-07	Issue units	TC-15	Planned	See SAD	See SAD
48	FR-REQ-08	Track and cancel	TC-30	Planned	See SAD	See SAD
49	FR-RPT-01	Stock report	TC-10	Planned	See SAD	See SAD
50	FR-RPT-02	Donation report	TC-32	Planned	See SAD	See SAD
51	FR-RPT-03	Request report	TC-32	Planned	See SAD	See SAD
52	FR-RPT-04	Filter reports	TC-32	Planned	See SAD	See SAD
53	FR-RPT-05	Export reports	TC-10	Planned	See SAD	See SAD
54	FR-RPT-06	Report access and empty result	TC-09	Planned	See SAD	See SAD
55	NFR-PERF-01	Search response time	TC-17	Planned	See SAD	See SAD
56	NFR-PERF-02	Page load time	TC-19	Planned	See SAD	See SAD
57	NFR-PERF-03	Concurrent users	TC-17	Planned	See SAD	See SAD
58	NFR-PERF-04	Report generation time	TC-32	Planned	See SAD	See SAD
59	NFR-SAF-01	No unsafe issue	TC-06, TC-15	Planned	See SAD	See SAD
60	NFR-SAF-02	Confirm group on issue	TC-15	Planned	See SAD	See SAD
61	NFR-SAF-03	Daily backup	TC-34	Planned	See SAD	See SAD
62	NFR-SEC-01	Password policy and hashing	TC-01	Planned	See SAD	See SAD
63	NFR-SEC-02	Role-based access control	TC-09	Planned	See SAD	See SAD
64	NFR-SEC-03	Encrypted transport	TC-35	Planned	See SAD	See SAD
65	NFR-SEC-04	Session security	TC-18	Planned	See SAD	See SAD
66	NFR-SEC-05	Input validation	TC-16	Planned	See SAD	See SAD
67	NFR-SEC-06	Audit trail	TC-21	Planned	See SAD	See SAD
68	NFR-SEC-07	No personal data in logs	TC-35	Planned	See SAD	See SAD
69	NFR-QUAL-01	Usability	TC-36	Planned	See SAD	See SAD
70	NFR-QUAL-02	Availability	TC-37	Planned	See SAD	See SAD

Sl.	Requirement ID	Description	Test case ID	Test status	Architecture ref	Design ref
71	NFR-QUAL-03	Data consistency	TC-33	Planned	See SAD	See SAD
72	NFR-QUAL-04	Maintainability and testability	TC-37	Planned	See SAD	See SAD
73	NFR-QUAL-05	Browser compatibility	TC-36	Planned	See SAD	See SAD
74	BR-01	Only Staff add donors/donations; only Admin manages staff and roles	TC-21, TC-22	Planned	See SAD	See SAD
75	BR-02	Users see only their own requests / donation history	TC-24, TC-30	Planned	See SAD	See SAD
76	BR-03	Ineligible donor cannot donate until next eligible date	TC-03, TC-27	Planned	See SAD	See SAD
77	BR-04	Earliest-expiry unit issued first (FEFO)	TC-08	Planned	See SAD	See SAD
78	BR-05	Urgent requests processed before Normal	TC-14	Planned	See SAD	See SAD
79	BR-06	Only Staff approve or reject requests	TC-31	Planned	See SAD	See SAD

14. Test Metrics & Reporting

Metrics collected:

- % test cases executed = executed / total x 100
- % passed and % failed = passed (or failed) / executed x 100
- Defect density = total defects / number of requirements or modules tested
- Defect aging = days between defect logged and closed
- Requirement coverage = requirements with at least one executed test case / total requirements x 100

Reports:

- Daily execution status (shared in the team chat)
- Final Test Summary Report

15. Approvals

Role	Name	Signature / Date
QA Lead	Dheepak Seetharam	04-10-2026
Dev Lead	Gautam Prasanth Nair	05-10-2026
Product Owner	Harsimran Kaur	06-10-2026